

Unified Consciousness Theory I — Publication Supplement

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Status: Publication supplement; not peer reviewed.

1. Publication posture

- English foundational theory preprint.
- Human author of record and responsible depositor: Hongju Liu.
- Substantial ChatGPT (OpenAI GPT-5.6 Sol) assistance under human direction is disclosed.
- This is adjacent first-party research and is non-amending with respect to the Trinity Accord and its three Bitcoin Originals.
- DOI registration, OTS, Arweave preservation, or public read-back establish version identity and availability; they do not establish truth, originality, peer review, significance, or indexing.
- The present package does not claim empirical validation of Universal Experience (A1) or the Experience-Structure Identity (A5).

2. Main contribution

Paper I changes the target of a universal consciousness theory under A1.

If every actual valid physical process token has non-empty experience,

$$E(P) = 1 \quad \forall P_{\text{actual}},$$

then a binary conscious/nonconscious classifier is constant on the UCT actual-process domain. The nontrivial scientific target becomes:

$$P \mapsto [\mathfrak{D}^{PA}(P)]_{\cong},$$

the physically anchored experiential organization of the process.

The paper further develops:

- a plural process-token ontology;
- the Experience-Subject Separation Principle;
- the Physically Anchored Experience-Structure Identity;
- the Physical Anchoring Condition;
- the Organization-Gate Impossibility result;
- chart covariance and multiscale plurality;
- a prospective empirical bridge program E1-E4.

3. Claim status

Axioms / foundational commitments

- A1 Universal Experience.
- A2 Continuity.
- A3 Nested Non-Exclusion.
- A4 Human Self Formation boundary.
- A5 Physically Anchored Experience-Structure Identity, including phenomenal completeness.
- A6 Token / Type / Lineage Distinction.

Conditional formal consequences

- Organization-Gate Impossibility under A1.
- Binary universality trivialization under A1.
- Chart covariance under structure-preserving invertible reparameterization.
- No-Hidden-Quale consequence under A5 completeness.
- Non-completeness of the tested scalar descriptor family by explicit counterexample.

Not established

- empirical truth of A1 or A5;
- historical priority;
- a complete solution to subject combination;
- a complete naturalized semantics of representation;
- R4 empirical validation;
- a solution to the hard problem.

4. Closest prior-work boundary

The paper does **not** claim novelty for any of the following in isolation:

- panexperientialism or panpsychism;

- intrinsic causal / structural approaches to consciousness;
- multiscale consciousness;
- overlapping conscious processes or systems;
- universalization of consciousness-theory constructs;
- process/coarse-graining methods;
- robust physical implementation constraints.

Relevant close pressure points discussed in the manuscript include IIT 4.0, Information Closure Theory, TICL, neurophenomenal structuralism, universal-theory criteria, overlapping-conscious-system arguments, robust-mapping/mechanistic computation, the Unfolding/Substitution debate, the combination/de-combination problem, and recent criticism of metaphysical structuralism.

The narrower candidate contribution is the conjunction of:

universal process-token experience+physically anchored structural identity+nested non-exclu

Historical-first priority is not claimed.

5. Empirical bridge program

Paper I proposes four prospective test classes:

1. **E1 — Phenomenal-geometry correspondence:** compare independently measured phenomenal similarity/difference structure with physically anchored response geometry.
2. **E2 — Trajectory-matched causal divergence:** compare systems/states with similar observable trajectories but different internal counterfactual response structures.
3. **E3 — Graded-perturbation continuity:** test continuous structural change across conditions where access/report observables show sharp thresholds.
4. **E4 — Multiscale macroprocess prediction:** test whether independently justified macroprocesses show stable causal organization without deleting lower-scale process descriptions.

These are prospective bridge tests, not completed validation.

6. Adversarial-review record

Before this prepublication freeze, Paper I underwent internal hostile-review simulations from four directions:

- panpsychism / combination and de-combination;
- metaphysical structuralism;

- physical computation / implementation;
- mainstream empirical consciousness science.

The review led to substantive revisions including:

- experience-bearing process $\neq$ canonical subject;
- A5 unpacked into structural correspondence + phenomenal completeness;
- PAC clarified as anti-triviality rather than a unique-natural-grain theorem;
- explicit E1-E4 empirical obligations;
- claim-language downgrade from unconditional “proof” to axiom-conditional derivation where appropriate.

These are internal reviews, not external peer review.

7. Numbering and repository boundary

This package reserves **TA-TR-2026-20** for Paper A.

The repository main research index currently lags the wider research sequence. **TA-TR-2026-19 must not be overwritten or reused.** Before merging Paper A into the public research index, reconcile the index so that the sequence remains explicit.

8. Release and preservation rule

This supplement is DOI-bound to the same version 1.0 record as the main paper.

The exact publication package must contain the reviewed DOI-bound files, citation metadata, and SHA-256 manifest. Publication is complete only after anonymous public exact-byte readback matches the reviewed manifest.

OTS submission begins only after the public Zenodo DOI package passes exact-byte readback. Bitcoin maturity must be verified before any guarded Arweave upload, and Arweave status is complete only after public readback passes.

Publication and preservation establish version identity and availability; they do not certify truth, peer review, originality, significance, or indexing.